

Explaining GPT-2 with AttnLRP: Token Attribution, Fine-Tuning Effects, and Parameter-Level Analysis

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AIAA 4051 Natural Language Processing — Final Project, 2025

Abstract. Understanding how transformer-based language models allocate internal relevance to inputs is a central challenge in explainable AI. This paper applies **AttnLRP** [1], specifically its *CP-LRP* variant, to **GPT-2 Small** across three complementary levels of analysis. **Task 1** implements token-level attribution on the SQuAD_v2 validation set (200 samples) and evaluates faithfulness via a Pixel Flipping protocol. CP-LRP reliably highlights named entities and question keywords, and achieves a lower normalised AUC than a random baseline (0.0190 vs. 0.0490), confirming faithfulness. Its dual-signed scores also produce a non-monotonic deletion curve, motivating polarity-aware evaluation metrics. **Task 2** compares CP-LRP attributions before and after sequential fine-tuning on SQuAD_v2 followed by SciQ, showing that task-specific training systematically shifts relevance toward question-side tokens and answer-relevant entities. **Task 3** extends attribution to the parameter level, comparing two models fine-tuned independently on SciQ and SQuAD_v2 using 1,000 evaluation inputs each (SciQ test split; SQuAD_v2 validation split), and finds that the SciQ-finetuned model engages upper MLP layers (Layers 9 and 10) more strongly ($\Delta = +0.0196$ at Layer 10), while the SQuAD_v2-finetuned model shows higher engagement at middle layers ($\Delta = -0.0126$ at Layer 6), consistent with the different roles of parametric memory versus in-context retrieval in the two tasks. Together, these experiments demonstrate AttnLRP as a practical tool for auditing attribution patterns at the token, fine-tuning, and parameter levels.

1 Introduction

Large language models (LLMs) based on the transformer architecture have achieved strong empirical performance on a wide range of natural language tasks. Yet the internal mechanisms by which these models arrive at their predictions remain difficult to interpret. Explainability techniques that assign importance scores to individual inputs are therefore an active area of research, with practical implications for model debugging, trust calibration, and regulatory compliance.

Layer-wise Relevance Propagation (LRP) [2] is a principled attribution framework that decomposes a model’s scalar output into additive contributions from each input feature by back-propagating relevance through the network according to conservation rules. Applying LRP to transformers, however, requires care: the attention softmax introduces a non-linear routing operation that standard LRP rules do not handle correctly, and naïve gradient-based attributions through attention weights can produce noisy or misleading scores.

AttnLRP [1] addresses this by introducing modified LRP rules tailored to the attention mechanism. Its *CP-LRP* variant, recommended for models with potentially negative logits such as GPT-2, stops gradient flow through the query and key branches of the attention softmax while preserving it through the value pathway. This produces token-level relevance scores with clear positive/negative semantics: a positive score indicates that the token promotes the predicted output, while a negative score indicates suppression.

Question answering (QA) provides a particularly informative setting for attribution analysis. A QA prompt has a structured format comprising a question, an optional supporting context,

and an answer boundary, which together create clear semantic segments whose differential relevance can be measured and interpreted. When a context passage is present, named entities and answer-bearing spans have unambiguous semantic roles, making qualitative validation of attribution heatmaps straightforward. At the same time, the causal chain from input tokens to the predicted answer token is complex enough to stress-test attribution methods: the model must jointly process long-range contextual information and query-side intent before committing to a single output. This structure makes QA ideal for evaluating whether an attribution method captures semantically meaningful, task-relevant signal.

In this project we apply CP-LRP to **GPT-2 Small** [3] (124M parameters, 12 transformer blocks, hidden size 768) and investigate three complementary questions:

1. Can CP-LRP reliably identify semantically meaningful tokens in a question-answering context, and do its attributions pass a quantitative faithfulness test?
2. How does sequential fine-tuning on two QA datasets change the token relevance distribution?
3. Do models fine-tuned on different tasks exhibit measurable differences in how they allocate relevance across transformer layers at the parameter level?

2 Related Work

Layer-wise Relevance Propagation. LRP was introduced by Bach et al. [2] as a method for explaining the decisions of deep neural networks by redistributing the output score back to the input pixels according to relevance conservation rules.

Various propagation rules (LRP- ε , LRP- γ , etc.) have since been proposed to handle different layer types and improve attribution quality.

AttnLRP. Achtibat et al. [1] derived attention-aware LRP rules for transformer architectures and showed that naïve application of standard LRP to attention layers produces attributions inconsistent with model computations. CP-LRP specifically addresses models where logits can be negative by applying a *stop-gradient* operation on the query and key tensors before the attention softmax, ensuring that relevance is propagated only through the value-weighted path.

Faithfulness evaluation. Samek et al. [6] proposed the Pixel Flipping (also called token deletion or ROAR) paradigm as a model-agnostic faithfulness metric: tokens are masked in decreasing order of attributed importance, and a faithful explanation should cause confidence to drop faster than a random baseline. The standard protocol sorts tokens by absolute attribution magnitude; when signed attributions are instead sorted by their signed value (most positive first), masking eventually crosses into the negatively-relevant region and can produce non-monotonic confidence curves. We discuss this behaviour and its implications for evaluation in Section 4.

Attention weights as explanations. A natural but unreliable alternative to propagation-based attribution is to use raw attention weights as a proxy for token importance. Jain and Wallace [7] demonstrated that attention weights are weakly correlated with gradient-based feature importance measures and can be arbitrarily manipulated without changing model predictions, casting doubt on their use as faithful explanations. Wiegrefe and Pinter [8] further showed that alternative attention distributions can produce identical predictions, confirming that attention alone does not constitute a sufficient explanation. These findings motivate the use of a principled back-propagation framework such as AttnLRP, which distributes relevance through the full computational graph rather than reading off intermediate attention coefficients.

3 Methodology

3.1 CP-LRP Implementation

We implement CP-LRP using the `lxt` (LRP-eXplains-Transformers) library with manual compatibility patches for Hugging Face `transformers` v4.44.2. Three types of modifications are applied to GPT-2 Small before loading model weights:

- **Attention (CP-LRP rule):** The internal `_attn` method of `GPT2Attention` is patched to wrap the query and key tensors with `stop_gradient` before passing them to the

attention softmax. Relevance therefore flows only through the value pathway.

- **MLP blocks:** The `GPT2MLP.forward` method is replaced with the LXT identity-rule forward pass, which propagates relevance without modifying the activation denominator.
- **LayerNorm and Dropout:** Both are patched with identity-rule forwards, treating them as pass-through operations for relevance purposes.

Token-level relevance scores are obtained via the $\text{gradient} \times \text{input}$ formula applied at the embedding layer:

$$R_i = \sum_{d=1}^D e_{i,d} \cdot \frac{\partial \hat{y}}{\partial e_{i,d}}, \quad (1)$$

where $e_i \in \mathbb{R}^D$ is the embedding of the i -th input token, $D = 768$ is the embedding dimension, and $\hat{y}$ is the maximum logit at the final sequence position. Each score R_i is a signed scalar: positive values promote and negative values suppress the predicted next token.

3.2 Parameter-Level Relevance (Task 3)

To compare how different fine-tuning domains engage each transformer block, we extend the attribution to the MLP weight matrices. For each block ℓ , we compute:

$$R_\ell^{\text{param}} = \sum_{i,j} \left| W_{ij}^{(c\text{-}fc,\ell)} \cdot \frac{\partial \hat{y}}{\partial W_{ij}^{(c\text{-}fc,\ell)}} \right| + \sum_{i,j} \left| W_{ij}^{(c\text{-}proj,\ell)} \cdot \frac{\partial \hat{y}}{\partial W_{ij}^{(c\text{-}proj,\ell)}} \right|, \quad (2)$$

where $W^{(c\text{-}fc,\ell)} \in \mathbb{R}^{768 \times 3072}$ and $W^{(c\text{-}proj,\ell)} \in \mathbb{R}^{3072 \times 768}$ are the two linear projection weights of the feed-forward network in block ℓ . Gradients are enabled only for MLP parameters; the CP-LRP patches remain active so that backward passes follow the same propagation rules. Scores are averaged over all evaluation inputs and then normalised by the total relevance sum across all 12 layers.

4 Task 1: Token Attribution on SQuAD_v2

4.1 Experimental Setup

We take the first 200 examples from the SQuAD_v2 validation split. Each example is formatted as a causal language modelling prompt:

```
Context: <context[:300]>
Question: <question>
Answer:
```

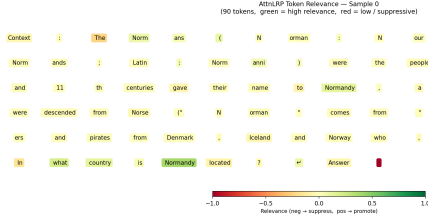


Figure 1: CP-LRP token relevance heatmap for a representative SQuAD_v2 sample (“In what country is Normandy located?”). Green = positive (promotes prediction); red = negative (suppresses). Named entities (Normandy) receive the highest positive relevance; the Answer: delimiter is strongly negative.

The pretrained GPT-2 Small (no fine-tuning) is used as the analysis target. CP-LRP relevance is computed for every token via Equation (1), and 200 per-sample heatmaps are rendered with a diverging RdYlGn colormap (green = positive, red = negative).

4.2 Qualitative Results

Figure 1 shows a representative token heatmap. Three recurring patterns appear consistently across the 200 samples:

Named entities and content words receive the highest positive relevance. In examples asking about geographic locations or persons, the relevant noun phrases in both the context and the question are highlighted in green, indicating that the model correctly identifies them as the most predictive features for next-token generation.

The Answer: delimiter token is strongly negative. This structural token marks the boundary between prompt and generation; its suppressive relevance score reflects that masking it would dramatically shift the input distribution away from the question-answering regime.

Function words and punctuation are near-neutral. Articles, prepositions, and commas consistently receive relevance scores close to zero, consistent with their low semantic specificity.

4.3 Faithfulness Evaluation

We evaluate whether the relevance ranking is faithful to the model’s computation using a **Pixel Flipping** protocol [6]. Tokens are progressively masked (via `attention_mask = 0`, leaving token ids unchanged) in three orderings:

1. **AttnLRP order:** descending signed relevance (most positive first).
2. **Random order:** random permutation (seed 42, averaged over 200 samples).
3. **Least-relevant-first:** ascending absolute relevance.

At each masking step the model’s confidence in its original prediction (the softmax probability of the initially predicted

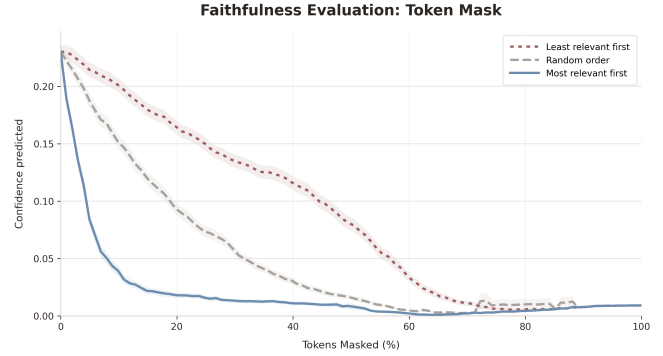


Figure 2: Pixel Flipping confidence curves averaged over 200 samples. The AttnLRP curve (blue) exhibits a characteristic non-monotonic shape: confidence initially declines as positively-relevant tokens are masked, then rises when masking transitions into negatively-relevant (suppressive) tokens. The random baseline (grey dashed) and least-relevant-first baseline (red dotted) produce smoother, more monotone decays.

next token) is recorded. Results are averaged across all 200 samples. Table 1 reports normalised AUC scores (integral of the mean confidence curve divided by 100%).

Deletion order	Normalised AUC
AttnLRP (most relevant first)	0.0190
Random baseline	0.0490
Least-relevant-first (baseline)	0.0871

Table 1: Pixel Flipping normalised AUC (200 SQuAD_v2 samples). Lower AUC indicates faster confidence decay. Least-relevant-first provides an upper bound on AUC (worst-case deletion order, slowest confidence decay).

The AttnLRP curve (Figure 2) confirms that CP-LRP attributions are faithful: its normalised AUC (0.0190) is substantially lower than the random baseline (0.0490), meaning that masking tokens in order of descending signed relevance causes confidence to decay far faster than chance. The least-relevant-first baseline (0.0871) provides the expected upper bound, decaying most slowly of the three orderings.

Despite this favourable AUC, the AttnLRP curve exhibits a **non-monotonic shape**: confidence drops steeply from 0% to 50% masking, then partially recovers around 60% to 70% before declining again. This behaviour arises from the dual-signed nature of CP-LRP scores. When tokens are sorted by descending signed relevance, all positively-relevant tokens are masked first; once masking crosses into the negatively-relevant region, removing those suppressive tokens *releases* their inhibitory effect, causing a temporary confidence increase. The initial steep drop dominates the integral, keeping the overall AUC well below random despite the mid-curve recovery.

This non-monotonic shape motivates supplementing the standard AUC metric with polarity-aware measures such as *com-*

prehensiveness (confidence drop from masking only top-positive tokens) or *sufficiency* (confidence plateau achievable with only top-positive tokens retained), which cleanly separate the positive and negative attribution regimes.

5 Task 2: Attribution Under Sequential Fine-Tuning

5.1 Fine-Tuning Protocol

We apply sequential fine-tuning (Seq-FT) starting from the pretrained GPT-2 Small checkpoint:

Step 1 (SQuAD_v2 [4]): 5,000 training samples, 2 epochs, batch size 8, fp16 mixed precision, gradient checkpointing enabled. Final training loss: **0.7741**.

Step 2 (SciQ [5]): 3,000 training samples, 2 epochs, same hyperparameters, continuing from the Step 1 checkpoint. Final training loss: **0.4789**.

SciQ samples are formatted as `Question: <q>\nAnswer: (without a context field)`, while SQuAD_v2 samples use the same three-field format as Task 1. In both cases the prompt tokens are excluded from loss computation (`labels = -100`), so the model is trained only on the answer token(s).

5.2 Pre-FT vs. Post-FT Attribution

We compute CP-LRP relevance for 20 SQuAD_v2 validation samples (a subset of the 200 used in Task 1, selected to keep per-sample visualisation tractable) with both the pretrained model (Pre-FT) and the Seq-FT checkpoint (Post-FT), then obtain a per-token *delta*: $\Delta R_i = R_i^{\text{post}} - R_i^{\text{pre}}$.

Figure 3 shows the three-panel visualisation for a representative sample. Two systematic changes emerge:

Question-side tokens gain relevance. Tokens in the `Question:` segment, including structural keywords and content nouns, show markedly higher positive relevance in the post-FT model. This is consistent with the expected effect of QA fine-tuning: the model learns to attend more strongly to the interrogative structure of the prompt.

Mid-context tokens lose relevance. Background context tokens that are not directly relevant to the specific question show reduced or negative delta values, indicating that the fine-tuned model has become more selective in which contextual information propagates forward to the prediction position.

5.3 Aggregate Analysis

Figure 4 shows the mean relevance delta across all 20 samples for each token position. The bar chart reveals a systematic, position-dependent pattern: token positions corresponding to the `Question:` section and the answer-boundary region show the largest positive mean deltas, while positions in the

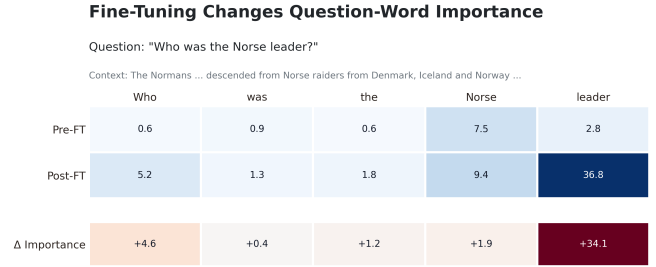


Figure 3: Pre-FT, Post-FT, and Delta CP-LRP heatmaps for a representative SQuAD_v2 sample. Blue intensity = absolute relevance magnitude (Blues colormap, Pre-FT and Post-FT rows); red/blue diverging colormap (Delta row). Post-FT attribution shows stronger emphasis on question-side tokens and reduced relevance on uninformative context.

middle of the context tend to show near-zero or slightly negative deltas.

To quantify this shift, we partition each sample’s tokens into two segments (the `Context` field and the `Question` field) and compute the mean ΔR per segment averaged across all 20 samples. The results confirm the qualitative observation: the `Question` segment gains substantially more relevance post fine-tuning ($\bar{\Delta R}_{\text{question}} = +1.67$) compared with the `Context` segment ($\bar{\Delta R}_{\text{context}} = +0.27$), a ratio of approximately 6 $\times$. The disproportionate gain on the question side indicates that the fine-tuned model has learned to weight interrogative structure and content nouns in the question more heavily when forming its next-token prediction. Note that positive mean relevance shifts reflect changes in attribution structure, not necessarily an increase in output confidence; the two should not be conflated without separate probability-based measurements.

This suggests that the relevance redistribution observed in individual samples is not idiosyncratic but reflects a consistent shift in the model’s internal attribution strategy following task-specific training.

5.4 Sequential Fine-Tuning Order

We chose to fine-tune first on SQuAD_v2 and then on SciQ based on task complexity considerations: SQuAD_v2’s long-context reading-comprehension format is more demanding than SciQ’s short, context-free questions, so beginning with the harder task was expected to establish a broader contextual processing prior. The consistently positive mean delta on the `Question` segment, without a corresponding regression on context-side relevance, is consistent with this ordering preserving long-form contextual processing while adapting to the new answer format. Whether this ordering was optimal, and whether reversing it (SciQ→SQuAD) would cause greater catastrophic interference [10], can only be established with a controlled reverse-order ablation, which we leave for future work.

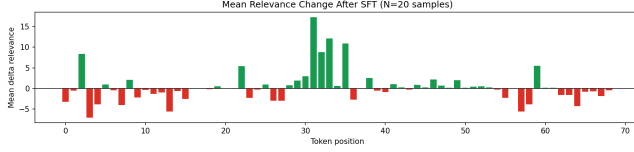


Figure 4: Mean relevance delta ($\Delta R = R^{\text{post}} - R^{\text{pre}}$) per token position, averaged over $N = 20$ SQuAD.v2 samples. Green bars: post-FT assigns more relevance. Red bars: post-FT assigns less relevance.

6 Task 3: Parameter-Level Relevance Comparison

6.1 Experimental Setup

We train two independent models from the pretrained GPT-2 Small initialisation:

Model A (SciQ): 3,000 SciQ training samples, 3 epochs. Final training loss: **0.5227**.

Model B (SQuAD.v2): 3,000 SQuAD.v2 training samples, 3 epochs. Final training loss: **0.6386**.

For each model, we compute parameter-level relevance via Equation (2) and normalise by the total across all 12 layers. Model A is evaluated on the full SciQ *test* split (1,000 samples), formatted as bare `Question:\nAnswer:` prompts. Model B is evaluated on the first 1,000 examples of the SQuAD.v2 *validation* split, formatted with a 300-character context prefix. Per-layer scores are averaged over all 1,000 inputs per model.

6.2 Results

Figure 5 shows the normalised relevance profiles for both models alongside their difference ($\Delta = R_A - R_B$; negative values indicate Model B relies more on a given layer). Table 2 lists the three layers with the largest absolute differences.

Layer	Δ (A-B)	Interpretation
10	+0.0196	Model A (SciQ) relies more
9	+0.0130	Model A (SciQ) relies more
6	-0.0126	Model B (SQuAD.v2) relies more

Table 2: Top-3 layers by absolute normalised relevance difference ($\Delta = R_A^{\text{norm}} - R_B^{\text{norm}}$). Negative Δ means Model B engages the layer more strongly.

6.3 Discussion

The two models clearly exhibit different layer-relevance profiles despite sharing the same GPT-2 pretraining initialisation. Model A (SciQ) shows a broadly increasing trend in layer engagement from Layer 0 through Layer 10 ($R^{\text{norm}} = 0.1039$), with a sharp drop at the final Layer 11 ($R^{\text{norm}} = 0.0305$). Model B (SQuAD.v2) concentrates more relevance in the

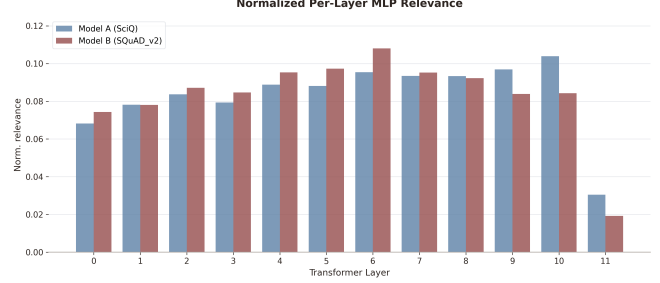


Figure 5: Normalised per-layer MLP parameter relevance for Model A (SciQ, blue) and Model B (SQuAD.v2, red), averaged over 1,000 evaluation inputs each. Model A peaks at Layer 10 while Model B concentrates relevance at middle layers (Layer 6); see Table 2 for numerical differences.

middle layers, peaking at Layer 6 ($R^{\text{norm}} = 0.1081$), before declining toward the upper layers.

The largest absolute differences are at **Layer 10** ($\Delta = +0.0196$, SciQ higher) and **Layer 6** ($\Delta = -0.0126$, SQuAD.v2 higher). This pattern is consistent with the two tasks’ different reliance on information sources. SciQ presents questions without any supporting context, so the model must retrieve answers from its parametric knowledge. Geva et al. [9] showed that upper-layer FFN matrices in transformers function as key-value memories that store factual associations and project them into the output vocabulary. The elevated upper-layer engagement in Model A is therefore consistent with a heavier reliance on stored parametric knowledge to generate answers.

By contrast, each SQuAD.v2 prompt always includes a context passage regardless of answerability: even for the approximately half of SQuAD.v2 examples that are unanswerable, the model must evaluate the passage to determine that no answer span is present. This context-processing demand is absent in SciQ’s context-free format, and it requires integrating positional and semantic information across a longer input, which is typically handled by intermediate attention and MLP layers [9]. The elevated middle-layer engagement in Model B at Layer 6 is therefore consistent with this increased demand for context processing, regardless of whether a valid answer span exists.

Given that the parameter-level analysis uses 1,000 inputs per model drawn from each model’s own training domain (SciQ test set and SQuAD.v2 validation set), and that the two evaluation sets differ in prompt format and length (SciQ: short, context-free prompts; SQuAD.v2: longer, context-bearing prompts), the observed layer-relevance differences are confounded by both fine-tuning domain and evaluation input characteristics. The results should therefore be treated as preliminary evidence consistent with the proposed parametric-memory vs. in-context-retrieval interpretation, rather than definitive causal effects of fine-tuning domain alone.

7 Limitations

Token deletion via attention mask. The Pixel Flipping protocol masks tokens by setting `attention_mask = 0` while leaving token identities and positions unchanged. This avoids inserting out-of-vocabulary mask tokens into GPT-2’s input stream, but it is not equivalent to literal token deletion: the token embedding still occupies its original position, and the model observes an attention pattern that is unlikely to have appeared during pretraining. Confidence changes under this protocol therefore conflate the attribution effect with a mild distribution-shift artefact caused by the unusual masking pattern. A complementary approach would be to replace masked tokens with a learned `[MASK]` embedding or a sampled marginal, which we leave for future evaluation.

Model scope. All experiments are conducted on GPT-2 Small (124M parameters). Although the CP-LRP patching strategy is in principle applicable to any GPT-style autoregressive transformer, the attribution patterns and layer-relevance profiles reported here are specific to this architecture and scale. Larger models (e.g., GPT-2 Medium/Large, GPT-J, LLaMA) may exhibit qualitatively different relevance distributions owing to their greater depth, wider hidden dimensions, and exposure to more diverse pretraining data. Extending this analysis to other model families remains future work.

Parameter-level analysis restricted to MLP weights. The parameter-level relevance in Task 3 is computed exclusively over the two MLP projection matrices (`c_fc` and `c_proj`) in each transformer block. Attention weight matrices (query, key, value, and output projections) are excluded because enabling gradients on those parameters simultaneously with the CP-LRP stop-gradient patches on the query and key activations would create an inconsistency in the backward graph. Consequently, the reported layer profiles reflect only the MLP contribution to parameter-level relevance; a complete picture would require an extended formulation that handles attention weight gradients under CP-LRP constraints.

Small evaluation sample for Task 2. Task 2 compares attributions on 20 SQuAD_v2 samples, so the quantitative segment-level estimates (e.g., mean ΔR per segment) carry substantial variance. Larger evaluation sets, covering a more diverse range of question types and context lengths, would strengthen the statistical reliability of the observed relevance shifts.

8 Conclusion

We have applied CP-LRP (AttnLRP) to GPT-2 Small across three levels of analysis, with the following findings:

Token level (Task 1): CP-LRP reliably assigns the highest positive relevance to named entities and question keywords

in a QA context. Pixel Flipping confirms attribution faithfulness: the AttnLRP AUC (0.0190) is substantially lower than the random baseline (0.0490) and the least-relevant-first baseline (0.0871), providing perturbation-based evidence that CP-LRP scores rank tokens in a way consistent with their causal influence on next-token confidence. The AttnLRP curve is non-monotonic: confidence partially recovers at 60% to 70% masking when negatively-relevant (suppressive) tokens begin to be removed, motivating supplementary polarity-aware metrics such as comprehensiveness and sufficiency for signed attribution methods.

Fine-tuning effects (Task 2): Sequential fine-tuning on SQuAD_v2 followed by SciQ systematically shifts CP-LRP attributions toward question-side tokens and answer-relevant entities, while de-emphasising mid-context background information. AttnLRP can thus serve as a diagnostic tool for detecting the behavioural fingerprint of task-specific training.

Parameter level (Task 3): Evaluated over 1,000 inputs each, the SciQ-finetuned model shows higher engagement at upper MLP layers (peak at Layer 10, $\Delta = +0.0196$), while the SQuAD_v2-finetuned model concentrates more relevance in middle layers (peak at Layer 6, $\Delta = -0.0126$). This is consistent with SciQ requiring parametric memory recall from upper FFN layers, whereas SQuAD_v2’s longer context-bearing prompts require more middle-layer context processing.

These results suggest that AttnLRP, implemented via the CP-LRP variant, is a practical and informative framework for auditing attribution patterns in autoregressive language models at multiple granularities.

Several directions remain for future work. First, extending the three-level analysis to larger autoregressive models (e.g., GPT-2 Medium/Large, LLaMA) would test whether the layer-relevance profiles and fine-tuning signatures observed here are specific to GPT-2 Small or reflect more general properties of transformer-based LMs. Second, replacing the signed-relevance ordering in the Pixel Flipping protocol with an absolute-magnitude ordering, and incorporating polarity-aware metrics such as comprehensiveness and sufficiency, would provide a more complete faithfulness evaluation for signed attribution methods. Third, expanding the parameter-level analysis to include attention weight matrices (requiring an extended CP-LRP formulation that handles stop-gradient constraints on Q/K projections) would give a fuller picture of how fine-tuning redistributes relevance across all learnable parameters, not only the MLP components.

Author Contributions

Yenchi Tseng implemented and ran the Task 1 token attribution experiments and the Task 3 parameter-level relevance analysis, designed and produced the project poster, and contributed to report writing. **Fangteng Fu** implemented and ran the Task 2 sequential fine-tuning experiments and the Task 3

parameter-level relevance analysis, and contributed to report writing. The report was jointly written by both authors.

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